Previously: A tree is a connected graph with no cycles. For any tree T we know $|E(T)| = |V(T)| - 1$.

Theorem (4.7). *If G is a connected graph, then* $|E(G)| \geq |V(G)| - 1$. $n = |V(G)|$, $m = |E(G)|$.

Not hard to see if $|V(G)| = 1, 2, 3$ Then $|E(G)| \geq |V(G)| - 1$





Assume not True.

So there is a graph G connected with $|E(G)| \leq |V(G)| - 2$ (m n)

Take such graph with minimal $|V(G)| = n$.

We know $n \geq 4$ and G contains no isolated vertices.

Claim: G contains a leaf. Assume otherwise $\deg v \geq 2 \; \forall v$.

$$2m = 2|E(G)| = \sum_{v \in V(G)} \deg v \geq \sum_{v \in V(G)} 2 = 2|V(G)| = 2n$$

$2m \geq 2n \Rightarrow m \geq n$ contradicts $n-2 \geq m$

Thus G contains a leaf v. Since $\deg v = 1$, Exercise. $G-v$ is also a connected graph.

But

$$|E(G \setminus v)| = |E(G)| - 1 \quad (v \text{ is a leaf, incident to unique edge})$$

$$|V(G \setminus v)| = |V(G)| - 1.$$

Thus $|E(G \setminus v)| = |E(G)| - 1 \leq |V(G)| - 2 - 1 = |V(G \setminus v)| - 2$

$\Rightarrow |E(G \setminus v)| \leq |V(G-v)| - 2$

Contradicts minimality of $n = |V(G)|$. !

$$\underbrace{|E(G)|}_{m} \leq \underbrace{|V(G)|}_{n} - 2$$

$\square$

Theorem (4.8). *If a graph* G *satisfies any two of the properties*

(1) G *is connected,* *(2)* G *contains no cycles.* *(3)* $|E(G)| = |V(G)| - 1$,

then G *is a tree.*

m n

Proof: (1)+(2) => Tree by definition.

(1)+(3): NTS G contains no cycles. Assume G contains a cycle, take an edge e in the cycle. Thus e is not a bridge. (Theorem 4.1). Thus G\e is connected (using (1))

But G\e has n vertices and $m-1 = n-1-1 = n-2$ edges

Contradicts previous Theorem.

(2)+(3) NTS G connected. Partition $G = G_1 \cup \dots \cup G_k$ into connected components. Now (corollary 4.6) G is a forest,

$|E(G)| = |V(G)| - k$ but (3): $|E(G)| = |V(G)| - 1 \Rightarrow k = 1$

$\Rightarrow$ G connected ☐

Theorem. *Let* T *be a tree on* k *vertices. If* G *is a graph with* $\delta(G) \geq k-1$, *then* T *is isomorphic to some subgraph of* G.

4.9

G contains a copy of T.

Proof: Induction on K. If $K=1$, $T = \bullet$ contained in any graph.

If $K=2$, $T = \bullet\!-\!\bullet$. If G has $\delta(G) \geq K-1 = 2-1 = 1$ Then G contains an edge, which is a copy of T.

Inductive hypothesis: If G is a graph with $\delta(G) \geq K-2$ Then it contains a copy of any Tree on $K-1$ vertices.

Let's show true for K. Take G graph with $\delta(G) \geq K-1$

Take a Tree T on K vertices,

Take a leaf v on T. Now $T-v$ is a tree on $K-1$ vertices.

Since $\delta(G) \geq K-1 \geq K-2$, by (IH), we know G

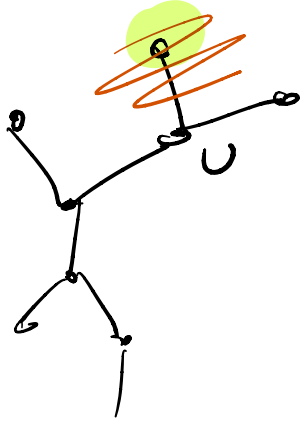
Contains a copy F of $T \setminus v$.

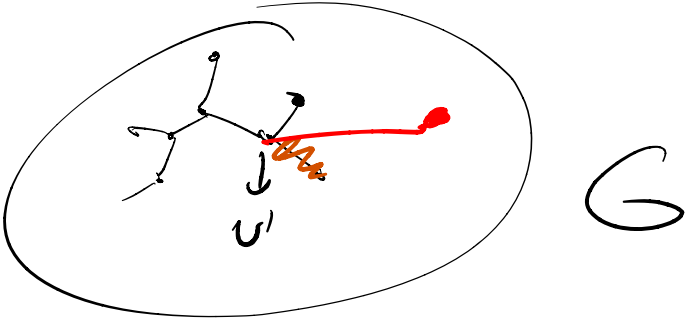
Let u denote the neighbor of v in T. Let u' denote it's copy on F. Note $\deg_G u' \geq \delta(G) \geq k-1$

Since F contains $k-1$ vertices ($k-2$ are different from u')

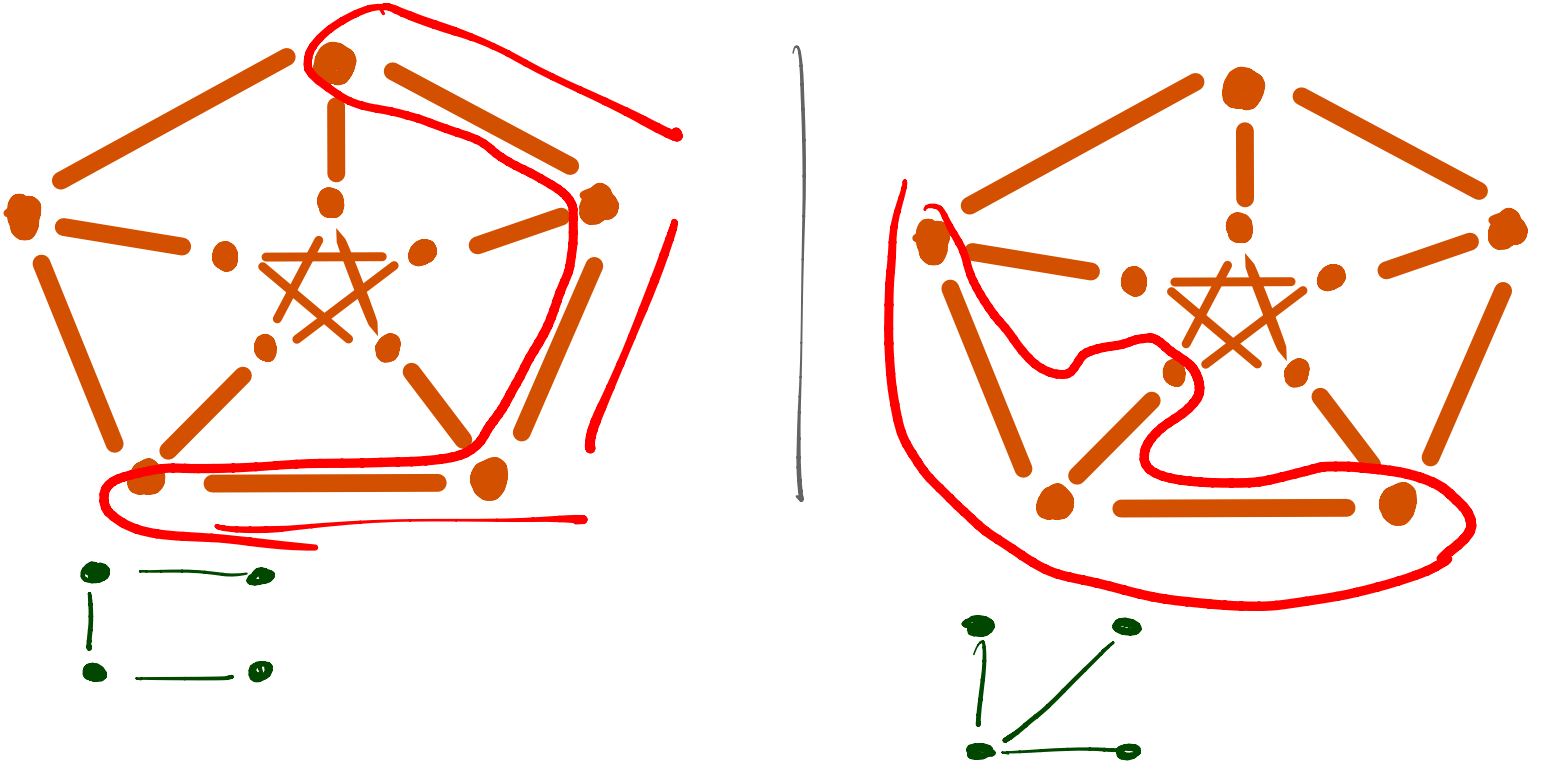
There is some $w \in G$ such that $u'w \in E(G)$.

So the graph obtained from F by adjoining w and $u'w$ is a copy of T inside of G.





Example. *Can you identify all trees on 4 vertices as subgraphs of the Peterson graph?* $\delta(G) = 3$



Exercise 1 (Corollary 4.6). *If a forest has exactly n vertices and k components, then it has $n - k$ edges.*

Exercise 2. *Show that every tree is bipartite.* Trees contain no cycles. (Theorem for bipartite)

Exercise 3. *Prove that a graph G is a tree if and only if G contains no cycle but $G + uv$ does contain a cycle for each pair of non-adjacent vertices u, v in G.*

Exercise 4. *Let T be a tree. For each $i \geqslant 1$, let n_i denote the number of vertices of degree i. Show that*

$$n_1 = 2 + n_3 + 2n_4 + 3n_5 + 4n_6 + \dots$$

components.

Take a forest G in n vertices with $G = G_1 \cup \dots \cup G_k$

Then $G_1, \dots, G_k$ are Trees → Connected ✓

→ Contain no cycles.

Thus $|E(G_i)| = |V(G_i)| - 1$. Then:

$$|E(G)| = \sum_{i=1}^{k} |E(G_i)| = \sum_{i=1}^{k} (|V(G_i)| - 1) = \sum_{i=1}^{k} |V(G_i)| - \sum_{i=1}^{k} 1$$

$$= |V(G)| - k \square$$

Exercise 1 (Corollary 4.6). *If a forest has exactly n vertices and k components, then it has $n - k$ edges.*

Exercise 2. *Show that every tree is bipartite.*

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$$n_1 = 2 + n_3 + 2n_4 + 3n_5 + 4n_6 + \dots$$

$|V(T)| = k$

n_k

$n_i = \#\{v \in V(T) : \deg v = i\}$.

$(n_1 \geq 2$ at least 2 leaves$)$

$$|E(T)| = |V(T)| - 1 = \sum_{i=1}^{k} n_i - 1$$

$$2|E(T)| = \sum_{v \in V(G)} \deg v = \sum_{i=1}^{k} i n_i$$